

From Air to Log-Mel — Solutions Manual

manim-concepts

Chapter 1

The spectrum: a bank of detectors

Practice problems

Problem 1.1. Eight samples are taken at 8000 per second. Here they are, above the 2000 Hz detector's two probes at the same 8 stops:

samples	5	12	-5	-12	5	12	-5	-12
cosine probe	1	0	-1	0	1	0	-1	0
sine probe	0	1	0	-1	0	1	0	-1

Compute the detector's pair (cosine sum, sine sum), its reading, and the amplitude of the tone; compare the amplitude with the largest sample. Then give the pair of the 1000 Hz detector, whose probes are $\cos(2\pi n/8)$ and $\sin(2\pi n/8)$ for $n = 0, \dots, 7$.

Solution. Cosine sum: $5 + 5 + 5 + 5 = 20$ (the -5 s meet the -1 s). Sine sum: $12 + 12 + 12 + 12 = 48$. The pair is $(20, 48)$ and the reading is $\sqrt{20^2 + 48^2} = 52$. On an interior row the amplitude is the reading $\div 4$: 13 — larger than any sample, the largest being 12, because the tone is $5 \cos + 12 \sin$ and peaks between stops (5–12–13). For the 1000 Hz detector, the samples satisfy $x[n+4] = x[n]$ while both 1-lap probes change sign four places later, so the terms cancel in pairs: $(0, 0)$.

Problem 1.2. All signals here are 8 samples at 8000 per second, read by the five-detector bank (0, 1000, 2000, 3000, 4000 Hz). (a) The signal is $2.5 \sin(2\pi \cdot 3000t + 50^\circ)$. Give all five readings. (b) The signal is $3 \sin(2\pi \cdot$

$1000t) + 0.5 \cos(2\pi \cdot 3000t)$. Give all five readings, and name the two facts that make the other rows read exactly 0.

Solution. (a) Only row 3 answers: $4 \times 2.5 = 10$, so the readings are 0, 0, 0, 10, 0; the 50° turns the pair, not its distance. (b) 0, 12, 0, 2, 0: row 1 reads 4×3 , row 3 reads 4×0.5 . The two facts: each probe's sum on a mix is the sum of its sums on the parts (a weighted sum is linear; the pair adds, and the distance is taken last), and every probe sums to exactly 0 against every other probe — all 28 pairs at $N = 8$ — so each tone reaches only its own row.

Problem 1.3. For each bank give the window's duration, the spacing between detectors, the number of rows, and the highest row's frequency. (a) Whisper's: $sr = 16\,000$ Hz, $N = 400$. (b) $sr = 48\,000$ Hz, $N = 960$. (c) You need detectors 10 Hz apart at $sr = 16\,000$ Hz: how many samples, and how long a window? If you switch to $sr = 48\,000$ Hz and keep that N , what is the spacing — and what N restores 10 Hz?

Solution. (a) $400/16\,000$ s = 25 ms; $16\,000/400 = 40$ Hz; 201 rows; up to 8000 Hz. (b) 20 ms; 50 Hz; 481 rows; up to 24 000 Hz. (c) $N = 16\,000/10 = 1600$ samples, 100 ms (801 rows). At 48 000 Hz the same N lasts a third as long and the spacing is $48\,000/1600 = 30$ Hz — coarser. 10 Hz needs $N = 4800$, which is again 100 ms: spacing is bought with time, at any sample rate.

Problem 1.4. Use $\log_{10} 2 \approx 0.301$. Levels are in dB relative to the loudest row unless a power ratio is named. (a) The loudest row reads 4 and another reads 0.5: what is the second row's level? (b) A row sits at -12.04 dB: what is its amplitude ratio to the loudest row, and the power ratio? (c) What is the level of a tenth of the amplitude? Of a tenth of the power? (d) A tone has half the *power* of the reference: what is its level, and its amplitude ratio? (e) The loudest row reads 4; another row sits at -26.02 dB. What does it read?

Solution. (a) $0.5/4 = 1/8$, three halvings: $-20 \times 3 \times 0.301 = -18.06$ dB. (b) $-12.04 = -20 \times 2 \times 0.301$, two halvings: an amplitude ratio of $1/4$, a power ratio of $1/16$. (c) $20 \log_{10} \frac{1}{10} = -20$ dB; for power, $10 \log_{10} \frac{1}{10} = -10$ dB. (d) $10 \log_{10} \frac{1}{2} = -3.01$ dB; amplitude goes as the square root of power, so the ratio is $1/\sqrt{2} \approx 0.707$. (e) $-26.02 = -20 - 6.02$: a tenth, then a half, so the ratio is $1/20$ and the row reads $4/20 = 0.2$.

Problem 1.5. (a) A 5000 Hz tone is sampled at 8000 per second. At what frequency is it heard? (b) At 16 000 per second, where are tones of 5000,

9000 and 10 000 Hz heard? (c) Write the 8 samples of $\sin(2\pi \cdot 7000 t)$ taken at 8000 per second, starting at $t = 0$, to two decimals. Which 1000 Hz tone has exactly these samples? Give the 1000 Hz detector's pair (cosine sum, sine sum) and reading. (d) The same for $\cos(2\pi \cdot 7000 t)$: pair and reading on the 1000 Hz detector.

Solution. (a) $8000 - 5000 = 3000$ Hz. (b) 5000 Hz is below $sr/2 = 8000$ Hz and stays put; $9000 \rightarrow 7000$ Hz and $10\,000 \rightarrow 6000$ Hz. (c) The samples are

$$0, -0.71, -1, -0.71, 0, 0.71, 1, 0.71,$$

those of $-\sin(2\pi \cdot 1000 t)$: a 1000 Hz sine turned upside down. The pair is $(0, -4)$ and the reading 4. (d) The samples equal those of $\cos(2\pi \cdot 1000 t)$ exactly: pair $(4, 0)$, reading 4. Either way the impostor is a clean 1000 Hz tone at full reading.

Problem 1.6. On the $N = 8$, $sr = 8000$ Hz grid the top row listens at 4000 Hz, exactly $sr/2$. (a) Write the samples of $\cos(2\pi \cdot 4000 t)$. What does row 4 read, and what divisor recovers the amplitude 1? (b) Write the samples of $\sin(2\pi \cdot 4000 t)$. What does every row read? (c) Write the samples of $3 \cos(2\pi \cdot 4000 t + 60^\circ)$. What does row 4 read, and what amplitude does the end-row divisor report?

Solution. (a) $1, -1, 1, -1, 1, -1, 1, -1$. Row 4's cosine probe is the same list, so it reads 8; the end rows divide by 8, not 4: $8/8 = 1$. (b) $\sin(\pi n) = 0$ at every stop: the samples are silence and every row reads 0 — which is why a tone must lie strictly below $sr/2$. (c) $3 \cos 60^\circ = 1.5$, so the samples are $1.5, -1.5, 1.5, \dots$ and row 4 reads $8 \times 1.5 = 12$; dividing by 8 reports 1.5, half the true amplitude. With no sine partner the end row reads $A \cdot N \cdot |\cos \varphi|$ — it depends on when the tone starts.

Problem 1.7. Exactly one of these five claims is true. Which — and what is wrong with each of the others? (a) “Between two samples a recording holds the earlier value, so the honest picture of 8 samples is a staircase.” (b) “8 samples at 8000 per second put the detectors 1000 Hz apart; 16 samples at 16 000 per second put them 500 Hz apart.” (c) “A row at 0 dB is silent.” (d) “A 7000 Hz tone sampled at 8000 per second shows up as noise spread across the rows.” (e) “Starting a 2000 Hz tone a quarter of a lap later changes the 2000 Hz detector's pair but not its reading.”

Solution. Only (e) is true: the pair turns by a quarter-turn and its distance stays put. (a) False: nothing between the samples is stored — a sample is a value at an instant, drawn as a stem. (b) False: both windows last

1 ms, so both banks are 1000 Hz apart; the faster rate brought reach (rows to 8000 Hz), and 500 Hz takes a 2 ms window. (c) False: 0 dB means equal to the reference ($\log 1 = 0$); silence is $-\infty$ dB. (d) False: it lands on 1000 Hz as a clean tone — row 1 reads 4 and every other row 0.

Problem 1.8. [stretch] The tone of the first problem starts one sample later, so the samples read $-12, 5, 12, -5, -12, 5, 12, -5$; two samples later they read $-5, -12, 5, 12, \dots$; three samples later $12, -5, -12, 5, \dots$. Give the 2000 Hz detector's pair (cosine sum, sine sum) and its reading at each of the three delays, and say what the sine probe alone would have reported at delays 0 to 3. Through what angle does one sample's delay turn the pair — and through what angle would it turn a 1000 Hz tone's pair on the 1000 Hz detector?

Solution. Delay 1: $(-48, 20)$. Delay 2: $(-20, -48)$. Delay 3: $(48, -20)$. Each is the previous pair turned a quarter-turn anticlockwise — $(a, b) \mapsto (-b, a)$ — and the reading is 52 every time. The sine probe alone reports 48, 20, -48 , -20 : the same tone, four different answers. One sample is $2000/8000$ of a lap, 90° ; for a 1000 Hz tone it is $1000/8000$ of a lap, 45° .